

**D-10-Y**

Roll No.....

Total No. of Questions : 16]

[Total No. of Printed Pages : 4

**12<sup>th</sup>ARM(H/Z)JKUTL2024-25****610-Y****BOTANY****(Biology)****Time : 1½ Hours]****[Maximum Marks : 35****SECTION-A****(OBJECTIVE TYPE QUESTIONS)****1 each****1. Which *one* is not decomposers ?**

(A) Fungi

(B) ~~Viruses~~

(C) Bacteria

(D) Flagellates especially abundant in the bottom of pond

**12<sup>th</sup>ARM(H/Z)JKUTL2024-25—610-Y****Turn Over****D-10-Y**

( 2 )

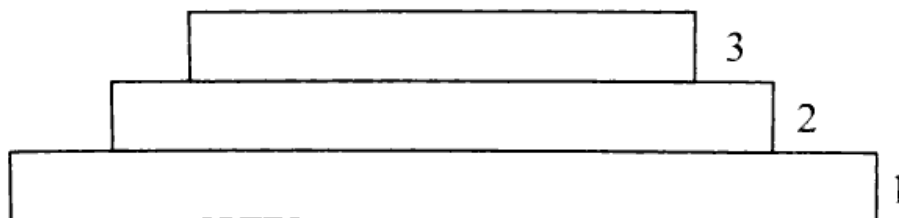
2. Large holes in Swiss Cheese are produced by :
- (A) A machine
  - (B) A bacterium that produces methane gas
  - (C) A bacterium that produces carbon dioxide gas
  - (D) A fungus that releases a lot of gases during its metabolic activities
3. A gene of operon which forms the repressor protein is :
- (A) Operator
  - (B) Promotor
  - (C) Regulator
  - (D) Structural
4. The technology of biogas production in India was :
- (A) Imported from Japan
  - (B) Imported from Denmark
  - (C) Imported from USA
  - (D) Developed in India due to the efforts of IARI and KVIC
5. Define Humification.

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( 3 )

**SECTION-B****(VERY SHORT ANSWER TYPE QUESTIONS)** 2 each

6. What is a Fermentor ?
7. A mature embryo sac in a flowering plant may possess 7-cells, but 8-nuclei. Explain it with the diagram only. <https://www.jkboseonline.com>
8. Label the three different tiers in the given age pyramid. Mention the kind of population growth described by the pyramid.



9. Define :
- (a) Pleiotropy
- (b) Test cross
10. Differentiate between integument and testa.

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( 4 )

**SECTION-C****(SHORT ANSWER TYPE QUESTIONS)**

3 each

11. Describe the development of monocot embryo.
12. Write a brief note on the importance of species diversity.
13. Describe the kind of food that would have lactic acid bacteria. Mention their useful applications.
14. What is Gene Expression ? Where would it be exerted in Eukaryotes ?
15. Differentiate between Dominance, Co-dominance and Incomplete dominance.

**SECTION-D****(LONG ANSWER TYPE QUESTIONS)**

5

16. What is Primary Productivity ? Give a brief description of factors that affect primary productivity.

*Or*

Define Population Interactions. Describe *four* types of negative interactions with the help of examples.

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**D-10-Y**

**B-10-X**

Roll No.....

Total No. of Questions : 16]

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**12<sup>th</sup>ARM(SZ)JKUT2024****1110-X****BOTANY****(Biology)****Time : 1½ Hours]****[Maximum Marks : 35****SECTION-A****1 each**

1. A diploid organism is heterozygous for 3 loci; how many types of gametes can be produced ?

(A) 3

(B) 6

(C) 8

(D) 12

**12<sup>th</sup>ARM(SZ)JKUT2024—1110-X****B-10-X****Turn Over**

( 2 )

2. Find the odd one :

(A) *Cephalosporium acremonium*

(B) *Acetobacter aceti*

(C) *Clostridium butylicum*

(D) *Lactobacillus*

3. Which one is not a biofertilizer ?

(A) *Azotobacter*

(B) *Bacillus thuringiensis*

(C) *Clostridium*

(D) *Azolla*

4. Which is an *in-situ* method of conservation ?

(A) Cryopreservation

(B) Biosphere Reserves

(C) Zoological Parks

(D) Wildlife Safari Parks

12<sup>th</sup>ARM(SZ)JKUT2024—1110—X

**B-10-X**

( 3 )

5. The second trophic level in a lake is :

(A) Phytoplankton

(B) Zooplankton

(C) Benthos

(D) Fishes

**SECTION-B**

2 each

6. Define Apomixis. Give its importance.

7. What is Double fertilization ?

8. Write a note on various types of processes that occur simultaneously during decomposition of detritus. <https://www.jkboseonline.com>

9. What is the key difference between Primary and Secondary sewage treatment ?

10. Draw schematic structure of a Transcription unit.

12<sup>th</sup>ARM(SZ)JKUT2024—1110—X

**B-10-X**

Turn Over

( 4 )

**SECTION-C**

3 each

11. What are Multiple Alleles ? Give their characteristics.
12. What is Tissue Culture ? Give its applications.
13. Flowers have evolved various strategies to prevent self pollination.  
Write a note on those strategies.
14. What are Hotspots ? Give an example and also write criteria for hotspots.
15. What are properties of genetic material ?

**SECTION-D**

5

16. Describe the attributes that populations possess but not individuals.

*Or*

Define Ecological Pyramid. Name the three types of ecological pyramids. What information is conveyed by each pyramid regarding structure, functions and energy in the ecosystem ?

12<sup>th</sup>ARM(SZ)JKUT2024

**B-10-X**



**B-23-X**

Roll No...

Total No. of Questions : **14**[Total No. of Printed Pages : **4****XIIARJKUT23****9123-X****BOTANY****(Biology)**Time : **1½ Hours****[Maximum Marks : 35****SECTION-A**

1 each

1. Control of gene expression takes place at the level of ;
  - (A) DNA replication
  - (B) Translation
  - (C) Transcription
  - (D) None of these
2. What is Cellular Totipotency ?
3. Name the organization that publishes the red list of species.
4. What is BOD ?

XIIARJKUT23-9123-X

**B-23-X**

Turn Over

( 2 )

**SECTION-B**

2 each

5. What do you understand by *in-situ* conservation ? Name some *in-situ* conservation strategies.
6. Describe the various parts of an orthotropous ovule.
7. What are Chasmogamous Flowers ? Why does a cleistogamous flower show autogamy only ?

*Or*

What is Anemophily ? What are the characteristics of anemophilous flowers ?

**SECTION-C**

3 each

8. What are the main causes of loss of Biodiversity ?
9. "DNA molecule is more stable genetic material than RNA." Explain, why ?
10. What are GMOs ? What are their pros and cons ?

( 3 )

11. What are Biocontrol Agents ? Giving examples show how they control pest infestation
12. Describe the development of a mature embryo sac from a megaspore mother cell in angiosperms.

*Or*

Describe the development of male gametes from microspore in flowering plants.

### SECTION-D

5 each

13. Inheritance pattern of flower colour in garden pea plant differs from that found in 4 O'clock plant. By showing crosses upto  $F_2$  generation, explain why this difference is observed ?

*Or*

What is semiconservative mode of DNA replication ? How was it experimentally proved and by whom ?

( 4 )

14. What are pioneer and climax communities ? Describe the process of ecological succession and differentiate between primary and secondary succession.

*Or*

What are Eltonian Pyramids ? With the help of one example each explain upright and inverted pyramids of number. Why is the pyramid of energy always upright ?

**A-3-A**

Roll No. \_\_\_\_\_

Total No. of Questions : 14]

[Total No. of Printed Pages : 3

**12<sup>th</sup>SZARJD22****6003-A****BOTANY****(Biology)**

Time : 1.15 Hours]

[Maximum Marks : 35

**Section-A**

1 each

1. \_\_\_\_\_ is called Anthesis. (Fill in the blank)
2. In angiosperms, functional megaspore develops into :  
(A) Endosperm  
(B) Pollen sac  
(C) Embryo sac  
(D) Ovule
3. The genotype of a plant showing a dominant phenotype can be determined by :  
(A) Back cross  
(B) Test cross  
(C) Dihybrid cross  
(D) Pedigree analysis
4. Name the chemical used for doubling the chromosome number.

12<sup>th</sup>SZARJD22-6003-A

Turn Over

**A-3-A**

( 2 )

**Section-B**

2 each

5. Differentiate between Autogamy and Xenogamy.
6. What is Cryopreservation ?
7. Differentiate between primary productivity and secondary productivity.

*Or*

Maximum number of trophic levels is hardly 3-5 in terrestrial ecosystems. Explain, why ?

**Section-C**

3 each

8. Describe the process of decomposition.
9. What are GM organisms ? What is their importance ?
10. Describe the importance of mycorrhiza, cyanobacteria and bacteria as biofertilizers. <https://www.jkboseonline.com>
11. With the help of well labelled diagrams explain the structure of an unfertilized and a fertilized embryo sac.
12. Define 'Genetic Code.' What are its characteristics ?

*Or*

Write explanatory note on the central dogma and central dogma reverse of molecular biology.

**Section-D**

5 each

13. What do you understand by Gene Expression ? Explain Lac Operon.

*Or*

Give an experimental verification of DNA as genetic material.

12<sup>th</sup>SZARJD22-6003-A**A-3-A**

( 3 )

14. Discuss the various types of interactions between the members of different species.

*Or*

Define Xerosere. Describe the process of ecological succession on a bare rock.

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12<sup>th</sup> SZARJD22-6003-A

**A-3-A**

<https://www.jkboseonline.com>

**B-3-B**

Roll No.....

Total No. of Questions : 14]

[Total No. of Printed Pages : 3

**XIIKDAR21****5003-B****BOTANY****(Biology)****Time : 1½ Hours]****[Maximum Marks : 35****Section-A**

1 each

1. ✓ RNA that pick up specific amino acid from cellular for protein synthesis is .....

(A) mRNA

(B) tRNA

(C) rRNA

(D) snRNA

(Choose one correct answer)

2. ✓ Removal of anther or stamens from flower bud to prevent self pollination is called ..... (Give one word answer)

**XIIKDAR21—5003-B****Turn Over****B-3-B**



( 2 )

✓ 3. What is triple fusion ?

4. Name *two* ODS.

**Section-B**

2 each

✓ 5. Write briefly advantages of micro-propagation.

6. What is Biofortification ? Write its *two* advantages.

✓ 7. What are Biopesticides ? Give some examples.

*Or*

Write some applications of tissue culture.

**Section-C**

3 each

✓ 8. How does the pollen mother cell develop into mature pollen grains ?

✓ Illustrate with the help of well labelled diagram.

✓ 9. Define Succession. Differentiate between Primary Succession and Secondary Succession.

*Or*

✓ What is the difference between a leading strand and a lagging strand ?

10. What is GM crop ? What are potential hazards of GM crops ?

**XIIKDAR21—5003—B****B-3-B**

11. Write *four* major factors regulating the size of a population.
12. What is Genetic Code ? What are its characteristics ?

*Or*

Write the applications of tissue culture.

**Section-D**

**5 each**

13. What is a dihybrid cross ? State and explain Mendel's Law of Independent Assortment with the help of a dihybrid cross.

*Or*

DNA acts as a 'Genetic Material.' Justify the statement with the help of an experimental evidence. <https://www.jkboseonline.com>

14. Enlist some threats to Biodiversity leading to its loss.

*Or*

What are the causes and effects of Air Pollution ? Write some control measures for air pollution.